

Milena Šošić

PhD in Computer Science | Machine learning & Artificial Intelligence | Natural Language Processing



GENERAL PROFILE

Machine Learning expert with 20+ years of experience in software engineering and machine learning algorithm design and implementation. Proven track record of delivering results in large-scale projects across finance, telecom, and commercial industries. Holds PhD degree in Computer Science, Natural Language Processing (NLP) specialization.

EDUCATION

Ph.D. in Computer Science (GPA 10/10)	2020–2025
University of Belgrade, Faculty of Mathematics	Belgrade, Serbia
Dissertation: <i>Modeling Moral and Emotional Language Aspects in Conversational Texts Classification</i>	
M.Phil. in Computer Science (GPA 10/10)	2004–2010
University of Belgrade, Faculty of Mathematics	Belgrade, Serbia
Thesis: <i>Applying classification methods on N-gram genome analysis</i>	
Diploma in Computer Science (GPA 8.86/10)	1997–2004
University of Belgrade, Faculty of Mathematics	Belgrade, Serbia

STUDY PROGRAMS:

Computer Science: Programming Languages (Pascal, C, C++, Java, Prolog, Lisp), Algorithms Architecture, Computer System Architecture, Mathematical Logic in Computer Science, Automata Theory, Databases, Microprocessors, Microcomputers, Assembly Language, Compilers and Interpreters, Programmable Systems Architecture.

Mathematics: Algebra (I and II), Analytical, Constructive and Euclidian Geometry, Calculus (I and II), Real and Complex Functions, Numerical Methods (I and II), Probability and Statistics.

TECHNICAL SKILLS

Programming: Python, C#, SQL

Architectures: RAG, GraphRAG, Agentic Frameworks

Databases: MS SQL Server, Oracle, IBM DB2, PostgreSQL, MongoDB, Redis, FAISS, Milvus, Azure Search Index, Vertex AI Search

Machine Learning and NLP: PyTorch, TensorFlow, HuggingFace Transformers, spaCy, nltk

LLMs and Tools: GPT, Llama, Qwen, Gemma, Unsloth, Promptflow, LangChain, LangGraph, DSPy, LangFuse, Ragas, DeepEval

Knowledge Graphs: RDF, OWL, SPARQL, Neo4j, Cypher

MLOps: Docker, Git, GitHub Actions, vLLM, LiteLLM

Cloud Platforms: Azure, GCP, AWS

Data and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Typesetting: $\LaTeX$

CERTIFICATES

Machine Learning	Jun, 2019
Stanford, Coursera	
Learn Python Programming Masterclass	Aug, 2019
Udemy	
Understanding and Visualizing Data with Python	Feb, 2020
MIT, Coursera	
Hands-On Machine Learning with scikit-learn and Python	Oct, 2020
Coursera	
Deep Learning Specialization (5 courses)	Oct, 2020
Coursera	
Natural Language Processing Specialization (4 courses)	Nov, 2020
Coursera	

PROFESSIONAL EXPERIENCE

Senior Data Scientist Technical lead for advanced AI and NLP solutions, including LLM-based systems, Retrieval-Augmented Generation (RAG) pipelines, agentic AI architectures, and knowledge graph solutions for enterprise applications.	Mar, 2024 –present Publicis Sapient
Senior Data Scientist Developed production-ready AI solutions with emphasis on NLP, predictive analytics, and applied research. Mentored junior data scientists and contributed to technical recruitment.	Jan, 2021 –Mar, 2024 Grid Dynamics
AI Team Lead Led the AI team and coordinated the development of intelligent software solutions. Introduced machine learning components into commercial products, with emphasis on text mining and sequence analysis.	Feb, 2019 –Jan, 2021 Devtech
Machine Learning Team Lead Led the development of machine learning solutions for Product Lifecycle Management (PLM), focusing on text classification, document processing, and business intelligence applications.	Jan, 2016 –Feb, 2019 Selerant
Data Scientist Developed predictive analytics models for customer segmentation, churn prediction, marketing, and risk assessment in the telecommunications domain.	Dec, 2013 –Jan, 2016 Telenor
Solutions Analyst Designed and optimized enterprise software solutions and business processes within SAP systems, supporting global business operations.	Dec, 2009 –Dec, 2013 PepsiCo
Senior Software Developer (.NET) Developed and maintained software systems for electronic voting using Microsoft .NET technologies.	Jan, 2007 –Dec, 2009 Dominion Voting Systems
Software Developer (.NET) Developed enterprise software solutions in C# and .NET, with emphasis on RFID technologies and industrial applications.	Aug, 2004 –Jan, 2007 SDD ITG

SELECTED PROJECTS

Hierarchical Multilabel Document Classification Selerant	Jan, 2017 –Feb, 2019
Semantic Query Parsing Grid Dynamics	Jan, 2021 –Jan, 2022
Email Classification into Business and Personal Categories Mathematics Faculty 	Jan, 2021 –Jun, 2022
Health-Care Search & Answer Publicis Sapient	Mar, 2024 –Apr, 2026
Agentic RAG for Technical Documents Search Publicis Sapient	Jun, 2025 –present

PUBLICATIONS & CONFERENCES

Effective Methods for Email Classification: Is it a Business or Personal Email? Computer Science and Information Systems 	Jun, 2022
SRPOL - A Lexicon Based Framework for Sentiment Strength of Serbian Texts Review of the National Center for Digitization 	Dec, 2022
Social-Emo.SR: Emotional Multi-Label Categorization of Conversational Messages from Social Networks X and Reddit Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig	Nov, 2024
Developing Language Resources for Recognizing Moral Aspects in the Serbian Language: Multi-label Categorization through the Perspective of Moral Foundations Theory Proceedings of the Artificial Intelligence Conference	Oct, 2025

Building an Emotion Lexicon for Serbian Using Curated Language Resources Language Resources and Evaluation, Springer 	Jan, 2026
Lexicon-Based Exploration of Emotion Dynamics in TEI Novels: Case Study on <i>SrpELTeC</i> Corpus Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig	Forthcoming
Hierarchical Recognition of Moral Foundations in Serbian Literary Corpus <i>SrpELTeC</i> Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig	Forthcoming
Exploring Narrative Frames via Emotional and Moral Signals in Text: A Knowledge Graph Approach to Analyze Social Discourse Proceedings of the Artificial Intelligence Conference	Forthcoming

AWARDS AND HONORS

Annual Award in Computing for Ph.D. Students Mathematical Institute of the Serbian Academy of Sciences and Arts	2026
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PROFESSIONAL SERVICE

Member of multidisciplinary academic association JeRTeh JeRTeh association is dedicated to development of language resources and technologies for Serbian language	2022–present Belgrade, Serbia
Advisory board member Serbian AI Society	2020–2022 Belgrade, Serbia

LANGUAGES

Serbian: Native
English: Full professional proficiency

PERSONAL SKILLS AND INTERESTS

Characteristics: Organized, detail-oriented, self-motivated and collaborative.
Hobbies: Hiking, reading, playing flute, traveling and exploring new technologies.